

# Design Principles for Exploratory Search Interfaces

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## Abstract

Exploratory search has been proposed as a model of search behaviour that is well suited to complex search scenarios. However, the simple interfaces that are commonplace across many search contexts limit the ability for searchers to undertake exploratory searches. Little support is provided for the discovery, learning, and investigation necessary for exploratory browsing, or the query (re)formulation, result examination, and information extraction required for focused searching. While the design and study of search interfaces that accommodate and support searchers in undertaking exploratory searches has increased in recent years, much of this work has been ad hoc in nature. In this perspective paper, five search interface design principles are presented that are specifically tuned to support exploratory search. An extension of the classical heuristic evaluation method is provided to support the inspection of prototype search interfaces with respect to the design principles. Recent research in the field is categorized according to these design principles. Patterns and gaps in the literature are identified, highlighting opportunities for further research on exploratory search interfaces. These principles provide a framework to guide the design and inspection of future search interfaces to support exploratory search, as well as a mechanism for comparing and contrasting the interactive information retrieval literature as it relates to supporting exploratory search through novel interface design.

## CCS Concepts

• **Human-centered computing** → *User interface design; Visualization design and evaluation methods*; • **Information systems** → *Search interfaces*.

## Keywords

search user interfaces, exploratory search, design principles, scoping review, heuristic evaluations

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## 1 Introduction

The “query box and ten blue links” design pattern has become the de facto standard in search interface design. It is well-suited to look-up search tasks, where a searcher is seeking specific and easily specified information or the verification of facts. However, not all information needs fall into this look-up category. In recent years, there has been great interest in studying how search tasks that are complex in nature might be supported through exploratory search [14, 25, 36, 37, 60, 66, 70, 74].

An information need is considered complex when it has one or more of the following aspects: open-ended, multi-faceted, ill-defined, or ambiguous. This, coupled with a searcher’s need to pursue the information need due to their lack of knowledge, create a complex search scenario. Such complex search scenarios can appear in a wide variety of situations, including but not limited to web search, academic or public digital library search, news search, and image/video search. A general pattern when undertaking a complex search scenario is that the information need cannot be satisfied by issuing a single query and selecting a single perfect document. As a result, searchers need to take an active role in the search process [21], engaging in search strategies that go beyond simple look-up behaviours, and be prepared for their search activities to span multiple search sessions.

Exploratory search has been proposed as a model of search behaviour that is well-suited to the pursuit of complex search scenarios [70]. This model suggests that searchers should start by seeking to learn broadly about the topic of interest through an exploratory browsing style of searching. As information is gained, confidence and knowledge about the search task increases, resulting in a transition to a targeted effort in finding relevant information through focused searching. This model does not describe a strictly linear process; it is possible that new facets of the information need are discovered in the focused searching activities, resulting in a decrease in certainty about the search task and the return to searching with a purpose of learning about the breadth of the topic [24]. As such search activities take time, the ability to complete the search task in a single session may not be possible [31].

How we engage in search processes is strongly influenced by the search interface at our disposal. For example, even when faced with complex search scenarios, searchers have a tendency to default to the simple search strategies that are promoted by simple search interfaces [25]. Clearly, there is a mismatch between the simple search interfaces that are commonplace, and the support needed for undertaking complex search scenarios using exploratory search processes. When undertaking an exploratory search, the simple search interface becomes a detriment, requiring the searcher to manage the complexity using other resources such as secondary documents, tabs or bookmarks, and even hand-written notes. Many

have argued for the need for better search interface support for exploratory search [21, 25, 70].

While many exploratory search interface designs have been proposed within the academic literature in recent years, most have been developed in an ad hoc way (including much of the author’s previous work). A common pattern is to identify an aspect of the exploratory search process that can be enhanced in some way, propose a new interface design or feature that is expected to better support the searcher, implement a prototype, and evaluate it in a user study. While this has resulted in a collection of approaches that support exploratory search processes in a variety of ways, it is difficult to make comparisons between the proposed approaches. There is a need for the interactive information retrieval research community to pursue a systematic approach to exploratory search interface design; a step forward is to identify principles that can guide the design process and be used to make comparisons between interfaces, which is the primary focus of this work.

The remainder of this paper is organized as follows. A review of existing design principles for search interfaces is presented in Section 2. New design principles for supporting exploratory search are provided in Section 3. In Section 4, a heuristic evaluation framework built around the design principles is presented to support the assessment and refinement of exploratory search interface prototypes. A scoping review is provided in Section 5, in which recent academic publications that present exploratory search interface designs are classified according to the design principles. The paper concludes with a discussion on the opportunities for further research and a call to action for further systematic development of search interfaces to support exploratory search across a broad range of search contexts.

## 2 Literature Review

There is a long history of developing interface design principles and guidelines within the field of human-computer interaction. Operating system vendors publish user interface guidelines, going back to the early days of Apple [57]. Their motivation was to encourage consistency in look and feel, enabling learnability as users switch between applications. As the web has matured, user interface guidelines have been developed in two directions: (1) accessibility guidelines [71] and (2) design patterns [55, 64]. These provide the foundational building blocks for high quality user interface design that transcends domains and contexts.

Within the field of information visualization, there is a similarly long history. Although Shneiderman’s Information Seeking Mantra (“overview first, zoom and filter, details on demand”) [56] is now more than 25 years old, it continues to serve as a popular guideline for information-centred user interface design. As it was presented with little evidence, some have analyzed the literature with respect to how Shneiderman’s information seeking mantra has been used, and have called for more formal design patterns to be developed [10]. Starting from a foundation in print media, Tufte has provided a popular set of design principles for visualizing quantitative information [62], with a strong emphasis on understandability and aesthetics. There are numerous textbooks on the topic [41, 67, 68], each providing their own flavour of design guidelines. More recently, there has been a move in the direction of

developing design patterns for specific information visualization contexts, such as dashboard design [2].

Although there are notably fewer works on the development of search interface design principles or guidelines, it is not without a foundation. More than 20 years ago, Bates published the Cascade Model, proposing that search interface designs should be informed by the search capabilities made available by the back-end information retrieval system and the activities and motivations of the searchers [3]. This model suggested that general principles for user interface design are not sufficient, and should be informed by the information retrieval context. On the heels of the rapid development and study of web search, site search, and digital library search interfaces in the 2000s, Hearst published a book on the current state-of-the-art in search user interfaces [20], including a series of design principles. Others have provided practitioner-focused guidance on the design of search interfaces and support for enhancing the search experience [40, 45, 54]. More recently, Liu et al. provided a survey of modern search interface design research, organized around four foundational stages in a search process (planning, searching, evaluation, and use) [32].

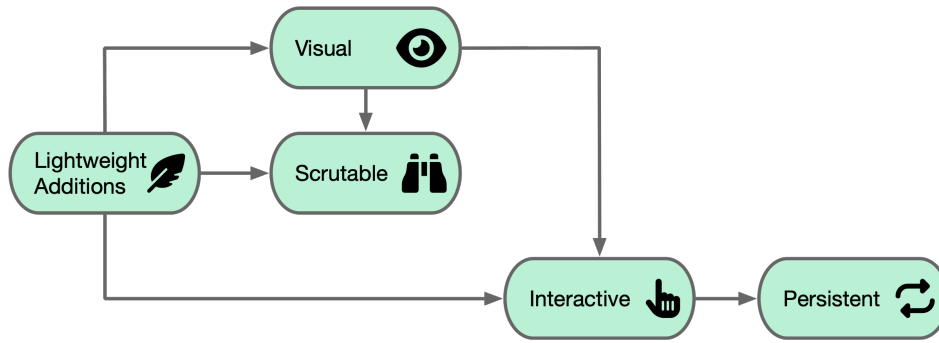
## 3 Design Principles for Exploratory Search Interfaces

Given that exploratory search processes require more support from the search interface than simple look-up strategies, generic design principles for interface design need to be augmented with specific design principles for supporting exploratory search. These design principles are informed by an awareness of the nature of complex search scenarios and the processes of undertaking exploratory searching [36, 60, 70]. The design principles are summarized in Figure 1 and described in detail below.

### 3.1 Lightweight Additions

As the “query box and ten blue links” design pattern permeates search interfaces across a wide variety of search contexts, significant deviations from this default design pattern will require that searchers learn how to use an entirely new search interface. Instead, *lightweight additions* can be used to enhance this design pattern rather than replace it. The ability to easily identify what the current query is and how to change it, and identify each of the search results in their ranked order are well supported with this default design pattern. Any new interface elements added to support exploratory search should be considered enhancements of this core interface design pattern, rather than replacements for it. Further, such additions should have a purpose for supporting specific aspects of exploratory search behaviour, rather than simply being added because metadata is available.

For example, consider a situation where it is imperative to show the relationship between search results in order to support the learning aspect of exploratory search (e.g., common authors, common sub-topics). A search interface designer might be tempted to create a graph-based (node-link) interface that connects and clusters related search results. Doing so would make it difficult for searchers to systematically assess the search results, and would make the underlying ranked order of the search results opaque. Another search interface designer may instead preserve the search



**Figure 1: The five design principles for exploratory search interfaces suggest the use of lightweight additions to the default search interface design pattern, the use of visual features that are scrutable, making these additions and visual features interactive, and persisting the interactive selections beyond the current search activity.**

results list paradigm as the core interface feature, and instead use other means to show the relationships between the search results, such as colour encoding or the use of glyphs. These other means would need to be lightweight additions made to each search result represented in the search interface so as to not overpower the core design pattern. Doing so would allow searchers to choose whether to use the list-based search interface as they normally would when pursuing simple look-up strategies, and to take advantage of the new lightweight features that show the relationships when their tasks dictate that an exploratory search process should be followed.

More tangible examples of lightweight additions to the default search interface design pattern that support exploratory search include faceted navigation controls [63], features that allow search results to be saved to workspaces [15, 66], and keyword summaries [24, 26, 37]. The key benefit in keeping the familiar search interface design pattern as the core of the search interface is that it enables the learnability of the new features by minimizing how much of the interface is actually new. Efforts to support learnability contribute to the overall usability of interfaces [17].

### 3.2 Visual

While a significant amount of information related to search is textual in nature, overloading a search interface with more and more information to read can make it difficult for searchers to decide where to focus their attention. The judicious use of visual encoding of some of this information can allow searchers to take advantage of their parallel visual information processing capabilities [68], enabling them to see the information without requiring the focused attention of reading. While historically, visual approaches used in search interfaces have not been well received [20], often such approaches either took a heavy hand in the application of visualization to the search interface (counter to the first design principle of lightweight additions), or added visual elements without consideration of the information seeking processes that needed to be supported. Even so, White & Roth’s foundational work on exploratory search suggested that visualization could be used to support insight and decision-making [70]. More recent work on the topic has suggested that there are opportunities to use visualization to enhance the evaluation of search results within exploratory search [32], and

that visual approaches can be particularly effective in showing patterns and relationships among information elements within search interfaces [22].

When considering how to apply visual approaches to a search interface, it is important to consider what aspect of the search process is to be enhanced. As a starting point, we can consider the fundamental search task stages of formulating a query, viewing search results lists, viewing content pages, and accessing bookmark pages [18]. Alternatively, the visual approaches can be applied in consideration of the exploratory search model such as enhancing exploratory browsing or focuses searching [70].

A secondary consideration is whether the visualization approaches are applied to the query, an individual search result item, or a collection of search result items such as the search engine results page (SERP) or a workspace. This not only determines the information source for the visualization, but also what aspect of the search interface will be affected. For example, using visualization to enhance the understanding of individual search results would require adding such visualizations to each result, whereas using visualization to enhance the understanding of a collection within a workspace would require adding the visualization to a place in the interface interpreted as representing the full set of search results (e.g., spanning the full width or height of the display).

Returning to the exploratory search model, there are opportunities for visual elements to be integrated into a search interface that supports the full breadth of associated activities. This includes the heuristic strategies that are part of the exploratory browsing style of searching: discovery, learning, and investigation; as well as the analytic strategies that are part of focused searching: query (re)formulation, result examination, and information extraction [24, 70].

Ultimately, the goal of information visualization applied to information retrieval contexts is to enable a searcher to perceive, interpret, and make sense of the information present in a search process [22]. To do this properly, the fundamental principles of information visualization must be applied, including the manipulation of visual variables given the data type [67] and the cognitive activity [6], and careful consideration of colour theory [1, 61], the Gestalt Principles [28], and data-to-ink ratios [12, 62]. Good visual representations within an exploratory search interface reduce

the need for the searcher to read information, which may reduce a searcher's cognitive load. Poor visual representations have the opposite effect, causing increases in cognitive load beyond what would have been required to read the information, due to searchers struggling to interpret and make sense of the visualization.

### 3.3 Scrutable

The concept of scrutability is closely linked to the concept of understandability, with the caveat that the item under consideration is being considered or examined (i.e., scrutinized) in the context of the task. As search interface designers add more and more information to support exploratory search processes (via lightweight additions that are visual), it is important to ensure that this information can be easily interpreted (matched to what the searcher knows) and made sense of (matched to the task being performed). As a result, ensuring that such search interfaces are scrutable is an important design principle that will ultimately contribute to a searcher's increased understanding of the topic as they conduct their exploratory searches.

Each new information element added to a search interface competes for the limited attention capacity of the searcher [19]. While we could consider the degree of scrutability of each element in isolation, consideration of the scrutability of the overall interface in a holistic manner is also important. Sometimes, the sum of the individual parts is not the same as the sum of the whole; that is, while each item in the interface might be decipherable and understandable, when provided together in a complete interface, the overall scrutability might diminish due to competing or conflicting signals in relation to the search task being performed.

As a result, the scrutability of a search interface becomes increasingly important to consider as more and more information is added to support exploratory search processes. The goal should be to develop interfaces that are simple in structure and organization, adding new elements to the interface only when necessary for supporting specific information seeking behaviours or activities needed for undertaking complex search scenarios. This is especially important with respect to the lightweight visual additions made to the core search interface, as it is not uncommon for interface designers to overdo these or to design them poorly, both of which lead to inscrutable presentations of information.

The concept of scrutability is linked to the sense-making stage of information visualization [22], as well as the crossing of the gulf of evaluation in human-computer interaction [44]. The ability to make sense of the information being presented in the interface is critical to being able to use it for the core task at hand. Likewise, when viewing information on a display, a user needs to be able to interpret and evaluate what they see as they plan their next action. What makes scrutability slightly different than these concepts is the willingness of the searcher to pay attention to and consider the element in relation to their specific exploratory search process and associated strategies.

### 3.4 Interactive

Even the simplest of search interfaces have fundamental interactive components of enabling searchers to enter their queries and select search results they wish to view. However, in order to support

exploratory search processes, there is a need to enable searchers to interact with the search process at a level that extends beyond such simple interactions.

Researchers have highlighted the importance of making search interfaces more interactive for decades [4, 5, 21]. As information objects are added to the search interface (e.g., via lightweight additions that are visual and scrutable), these become information objects that can be interacted with in support of undertaking an exploratory search. How best to support interactions with such objects remains an open challenge in interactive information retrieval research [4]. Such interaction can assist searchers in providing more information about what they are seeking beyond simply typing a query, and more information about what they find useful beyond simply clicking on a search result. For example, a hypothetical search interface could surface concepts extracted from a list of search results and allow the searcher to select or order these concepts based on their current interests; doing so could serve as a mechanism for learning about the relationships among the search results and discovering related content. Another hypothetical search interface could allow searchers to drag-and-drop search results into a set of pre-defined categories (e.g., save for now, save for later, discard) to structure the result examination process.

An important concept in interaction design is to ensure that every interactive element of the interface provides clear and understandable feedback [43, 57]. Doing so ensures that searchers remain aware of the outcomes of their interactions, contributing to their engagement in the exploratory search process. In the first hypothetical example above, the interaction of selecting or ordering of the concepts might visually show the outcomes of this interaction as well as animate a filtering or re-ranking of the search results to illustrate how the interaction affects the search results list. In the second hypothetical example above, the drag-and-drop of a search result into a category could remove the result from the search results list and update a counter for the number of items in the target category.

Although some have provided an extensive analysis of the different types of interactions that are possible in visually-intensive interfaces [73], the spirit of the scrutability design principle should be preserved here too. That is, the interaction should be clear, understandable, and obvious, enabling the searcher to easily consider what they can do with the interface, thereby enabling them to cross the gulf of execution [44]. Fundamental human-computer interaction design patterns should be considered, giving users visual cues that an element is interactive using metaphors to model interactive elements after real-world interactive objects such as buttons and folders [57] and de facto standards such as underlining clickable text or changing the cursor from a pointer to a finger when hovering over an interactive element that is clickable.

### 3.5 Persistent

Assuming that a search interface designed to support exploratory search includes interactive elements that allow a searcher to take an active role in the search process [21], consideration should be given as to whether this interactivity should be applied only to the immediate search activity or saved and applied to other related search activities. In some cases, the interactivity might only be

relevant to what the searcher is doing at that point in time; in other cases, it might be relevant to the overarching exploratory search process. In the later case, persisting the interactive selections may streamline the search activity by minimizing the need to perform repetitive interactions during the search session.

Consideration should also be given to persisting such interactions across search sessions. When faced with a complex search scenarios and undertaking exploratory search processes, studies have found that searchers often extend their searches across multiple sessions [31]. A critical aspect of any interface that is designed to support cross-session searching is providing mechanisms to support the searcher in re-acquainting themselves with what they have done thus far [15]. In addition to providing access to saved queries and associated resources, preserving the interactive state of the interface across search sessions is expected to further support task resumption as it will save the searcher from having to repeat the interactive selections made in the previous search session.

Consider once again the hypothetical example of providing concepts extracted from a set of search results and the interactivity of allowing the searcher to select relevant concepts to filter the search results list. The selection of a concept of interest might initially be done in the context of the current set of search results. However, if this concept is selected and a subsequent query is issued, perhaps there is value in having this same concept automatically selected for the new set of search results, as the searcher has already shown that it is something of interest. Further, it may be beneficial to remember this selected concept in a cross-session search scenario. Of course, the ability to easily remove previous selections or perform a full reset becomes an important feature in the case where a searcher has decided to explore a new facet of the complex search scenario or change their search topic entirely.

Making such interactions persist beyond the activity in which they were performed can allow a search interface to become personalized to the searcher's current search context. The scale of the persistence can be considered in the context of how long the search interface remembers the interaction: subsequent pages from the same query (result persistence), subsequent queries and corresponding search results or viewing the workspace in the same session (session persistence), or subsequent sessions when returning to the same complex search scenario (cross-session persistence). The choice among these options depends on whether the interaction is considered local to the immediate search activity (e.g., re-sort the current and next pages of the search results based on a specific attribute) or could be relevant to the overall exploratory search process (e.g., highlight the use of a specific term that has been learned as a valuable indicator of relevance across all searches on this topic, in the current session and future sessions).

Not all interactive elements of a search interface need to be persisted. Careful consideration should be given as to whether remembering the interaction might streamline the overall exploratory search process and enable the searchers to increase their understanding and certainty in their complex search scenarios. If during pilot testing or user studies of a prototype interface observations are made of searchers performing repeat actions, this could be an indication of the need for persistence being added to the search

interface design; likewise the use of reset functions or otherwise removing the previous selections that have been saved could be an indication of having taken the concept of persistence too far. Minimizing the need for users to have to repeat their interactive selections is well aligned with the fundamental user interface design principle of keeping users in control [57].

## 4 Heuristic Evaluations for Exploratory Search Interfaces

Heuristic evaluations are a common approach for debugging an interface following well-established usability guidelines. In foundational work on heuristic evaluations, nine usability heuristics served as a guide for a study that showed three to five assessors were able to identify most issues [43]. These nine heuristics have since expanded into ten [42], and have been extended to include heuristics that are specific to task domains [51] as well as heuristics for specific user interface design methods such as the use of information visualization techniques [75].

The design principles presented in this paper have been translated into a set of usability heuristics for exploratory search interfaces. These heuristics are intended to extend existing usability heuristics from the field of human-computer interaction, not replace them. That is, these heuristics should be applied in addition to the ten foundational usability heuristics [42].

The following exploratory search interface heuristics are presented in the same format as a common workbook for conducting heuristic evaluations [39, 42], providing a summary of the heuristic and a set of questions to be considered when evaluating the interface.

- (1) **Lightweight Additions:** Any additions to the search interface design should preserve the standard query box and search results list as the core design pattern for collecting the searcher's query and presenting the search results.
  - Does the design allow the searcher to easily identify where to enter or modify their query?
  - Are the search results organized to enable an interpretation of their ranked order?
  - Do the lightweight additions conflict with or detract from the core search interface design pattern?
- (2) **Visual Representations:** The search interface design should use visual representations of metadata in order to minimize reading and promote the identification of patterns or relationships.
  - Is the metadata that is being visualized useful for supporting discovery, learning, and investigation during exploratory browsing, or query (re)formulation, result examination, and information extraction during focused searching.
  - Does the design follow fundamental information visualization principles for the given data types and intended visual analysis?
  - Can patterns or relationships be inferred from the visual representations?

| Publication Venue                                                               | Harvested Results | Search                  |                    |
|---------------------------------------------------------------------------------|-------------------|-------------------------|--------------------|
|                                                                                 |                   | Interface Contributions | Exploratory Search |
| ACM SIGIR Conference on Human Information Interaction & Retrieval (CHIIR)       | 43                | 20                      | 13                 |
| ACM CHI Conference on Human Factors in Computing Systems (CHI)                  | 29                | 9                       | 2                  |
| ACM SIGIR Conference on Research & Development in Information Retrieval (SIGIR) | 27                | 8                       | 4                  |
| Journal of the Association for Information Science and Technology (JASIST)      | 12                | 3                       | 2                  |
| ACM Symposium on User Interface Software and Technology (UIST)                  | 7                 | 2                       | 2                  |
| Proceedings of the Association for Information Science & Technology (ASIS&T)    | 6                 | 0                       | 0                  |
| Information Processing & Management (IP&M)                                      | 5                 | 1                       | 1                  |
| ACM Conference on Intelligent User Interfaces (IUI)                             | 4                 | 1                       | 1                  |
| Total                                                                           | 133               | 45                      | 25                 |

**Table 1: Publication venues and descriptive statistics of the scoping review.**

- (3) **Scrutable:** With a small amount of focused attention, the search interface design should allow the searcher to make sense of the information being communicated.
  - Can the searcher make sense of the visual representation with respect to the activity that is to be supported (querying, results examination/comparison, workspace analysis, task resumption)?
  - Does scrutinizing an interface element have the potential to reduce uncertainty about the complex search scenario?
  - Is the amount of attention required a reasonable trade-off for the value provided for supporting exploratory browsing or focused searching?
- (4) **Interactive:** The search interface design should allow searchers to interact with the search process beyond entering a query and selecting search results.
  - Are visual cues or metaphors used to make it obvious how a searcher is expected to interact with the interface elements?
  - Is the interaction allowing the searcher to tell the interface more about their complex search scenario than simply typing in a query?
  - Does the interaction provide meaningful feedback that contributes to discovery, learning, and investigation during exploratory browsing, or query (re)formulation, result examination, and information extraction during focused searching.
- (5) **Persistent:** The search interface design should persist interactive selections such that they can be used to inform subsequent search activities.
  - Are the previous interactions remembered between search result page changes?
  - Are the previous interactions remembered when new queries are submitted in the same session?
  - Are the previous interactions remembered across multiple search sessions?

## 5 Scoping Review: Classifying Existing Approaches

In order to evaluate to what degree these design principles are already being used within recent search interface design research,

a scoping review was conducted and the resulting literature was analyzed with respect to these principles. This scoping review was conducted by the sole author of this article; as an experienced researcher in the field they are able to assess the literature accurately.

A harvest query of “search interface” OR “search user interface” was used, along with a filter that limited the search to works published in the last four years (2021 - 2024). The search was issued within the primary conferences and journals in which search interface innovations are expected to be published. Each publication was reviewed and assessed based on two criteria: (1) was it making a contribution to search interface design or innovation, and (2) did the authors make the case for the proposed interface to support exploratory search or a fundamental aspect of exploratory search (e.g., learning, investigating, etc.). Table 1 provides the list of these publication venues, along with the number of publications that matched the harvest query/publication date filter, and the number that fulfilled the screening criteria of making a search interface contribution and of those, supporting exploratory search. Although research on conversational search is becoming increasingly popular, our focus here is on tangible screen-based interfaces; we save the task of modifying these design principles to a conversational search context as future work for those working in this specialized aspect of search interface design.

The primary publication venue for novel search interface designs for complex search tasks over the last four years is CHIIR (13). This is as one might expect, as it is the premier publication venue for research at the intersection of information retrieval and human-computer interaction. Some work has also been published in the larger venues that represent these two different aspects of a search user interface: SIGIR (4) and CHI (2). Publications have also appeared in other related interface design conferences: UIST (2) and IUI (1). However, no relevant publications were found in the top conference in information science/library science: ASIS&T (0). The two leading journals in the field have published a small number of articles that are relevant for this scoping review: JASIST (2) and IP&M (1).

Table 2 provides an assessment of each publication that made a search interface contribution and was designed to support complex search tasks, with respect to the five design principles. The publications are sorted by year and first author; multiple publications on the same search interface are clustered, with one assessment made for all.

| First Author, Year    | Venue  | Lightweight Addition | Query           | Visual Representation      |                 |                            | Scrutability | Interactivity    | Persistent               | Context    |
|-----------------------|--------|----------------------|-----------------|----------------------------|-----------------|----------------------------|--------------|------------------|--------------------------|------------|
|                       |        |                      |                 | SERP                       | Result Item     | Workspace                  |              |                  |                          |            |
| Câmara, 2021 [7]      | CHIIR  | ✓                    |                 | colour encoding            |                 | hierarchy                  | ✓            |                  |                          | Web        |
| Choi, 2021 [9]        | CHIIR  |                      |                 |                            |                 |                            |              |                  |                          |            |
| Crescenzi, 2021 [11]  | SIGIR  | ✓                    |                 |                            |                 | nested cards               | ✓            | drag-and-drop    | global                   | News       |
| Li, Y., 2023 [30]     | JASIST |                      |                 |                            |                 |                            |              |                  |                          |            |
| Gomes, 2021 [16]      | CHIIR  | ✓                    | timeline        | highlighting               |                 | highlighting               | ✓            | selection        | results, session, global | Academic   |
| Gomes, 2022 [15]      | SIGIR  |                      |                 |                            |                 |                            |              |                  |                          |            |
| Li, X., 2021 [29]     | CHI    |                      |                 |                            |                 |                            | ✓            | selection        |                          | Scientific |
| Medlar, 2021 [37]     | CHIIR  | ✓                    | histogram       | node-link, colour encoding |                 |                            | ✓            |                  |                          | Academic   |
| Palani, 2021 [47]     | CHI    | ✓                    |                 |                            |                 |                            |              | note taking      | global                   | Web        |
| Shukla, 2021 [58]     | CHIIR  |                      |                 |                            |                 |                            |              |                  |                          |            |
| Hoerber, 2022 [24]    | JASIST | ✓                    |                 |                            | highlighting    | highlighting               | ✓            | selection        | results, session         | Academic   |
| Hoerber, 2024 [23]    | CHIIR  |                      |                 |                            |                 |                            |              |                  |                          |            |
| Kaur, 2022 [27]       | UIST   | ✓                    |                 | histogram                  | colour encoding |                            |              | selection        |                          | Academic   |
| Liu, Y., 2022 [33]    | SIGIR  | ✓                    | facets          |                            |                 |                            | ✓            | selection        |                          | Image      |
| Liu, Y., 2024 [34]    | IUI    |                      |                 |                            |                 |                            |              | selection        |                          | Medical    |
| Liu, Y.-H., 2022 [35] | CHIIR  | ✓                    |                 |                            |                 |                            |              | selection, lasoe | session                  | Web        |
| Palani, 2022 [48]     | UIST   | ✓                    |                 |                            |                 | linked cards               | ✓            |                  |                          |            |
| Ruotsalo, 2022 [53]   | IP&M   | ✓                    |                 |                            | colour encoding |                            | ✓            | selection        |                          | Movie      |
| Vuković, 2022 [65]    | SIGIR  | ✓                    |                 | timeline                   |                 |                            | ✓            | selection        |                          | Quotations |
| Chavula, 2023 [8]     | CHIIR  | ✓                    |                 |                            |                 | node-link                  | ✓            | drag-and-drop    |                          | Academic   |
| Payandeh, 2023 [49]   | CHIIR  | ✓                    | colour encoding |                            |                 | node-link, colour encoding | ✓            | drag-and-drop    | session, global          | Library    |
| Soufan, 2023 [59]     | CHIIR  | ✓                    |                 | concept map                |                 |                            | ✓            |                  |                          | Academic   |
| Mishra, 2024 [38]     | CHIIR  |                      | sliders         |                            | colour encoding |                            |              |                  | global                   | News       |
| Pirmoradi, 2024 [50]  | CHIIR  | ✓                    |                 | highlighting               | tags            | highlighting               | ✓            | selection        | session                  | Academic   |
| Reed, 2024 [50]       | CHIIR  | ✓                    | facets          |                            |                 |                            | ✓            | selection        | results                  | Web        |

**Table 2: A summary of academic publications on search interfaces for complex search tasks in relation to the five design principles for exploratory search interfaces. Publications are ordered by year; those represent the same search interface discussed in multiple venues are grouped in the same row.**

| Visualization Method | Description                                                                                                                            | Used In                  |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| colour encoding      | Using colour to encode qualitative or quantitative attributes.                                                                         | [7, 27, 29, 38, 49, 53]  |
| hierarchy            | Using structure or lines to show parent-child relationships.                                                                           | [7]                      |
| nested cards         | Using cards to represent entities, and nesting these within one another to show parent-child relationships.                            | [9, 11, 30]              |
| timeline             | Representing temporal aspects on a line graph; using other visualization methods such as size or colour to represent other attributes. | [15, 16, 65]             |
| highlighting         | Marking all elements of the same type is a common visual attribute such as colour or with a specific mark.                             | [15, 16, 23, 24, 50, 58] |
| node-link            | Representing entities as nodes and using lines to link the entities in order to show their relationship.                               | [8, 29, 49]              |
| histogram            | Representing the frequency of the entities use the length of a bar, organized on a common axis.                                        | [27, 37]                 |
| facets               | Structured representation of metadata of the entities, listing the attributes for each facet.                                          | [33, 34, 52]             |
| linked cards         | Using cards to represent entities, and lines to link the entities in order to show their relationship.                                 | [48]                     |
| concept map          | A specific type of node-link diagram that represents topic concepts and their relationships with one another.                          | [59]                     |
| sliders              | Representing the values of attributes on parallel fixed-length axes using visual elements that look like physical sliders.             | [38]                     |
| tags                 | Simple shapes (e.g., ovals) with textual labels representing user or system generated attributes of the entities.                      | [50]                     |

**Table 3: Brief descriptions of the visualization methods used in the exploratory search interfaces.**

The lightweight addition principle was considered as a binary decision (either the search interface preserved the standard query box and search results list as the core of the interface, or it did not). The visual representation principle was considered with respect to whether the visualization was applied to the query or in support of the query process, for the full search engine results page (SERP), an individual search result item, or as part of a workspace; the type of visualization was identified. The scrutability principle was considered as a binary decision as to whether it would be feasible for a searcher to make sense of the interface in relation to their search task. The interactivity principle was considered in relation to the proposed search interface innovation, with the type of interactivity documented. The persistence principle was considered with respect to how long the interactivity persisted (within additional pages of results for a query, within multiple queries over a session, or globally across multiple sessions). Finally, the search context for each interface is documented.

Many of the search interface publications in this scoping review preserved the “query box and ten blue links” search interface design pattern, adding new features as dictated by their search context and specific goals for enhancing the exploratory search process. Of note, twelve of the thirteen publications from CHIIR are aligned with the lightweight additions principle; one of the four publications from SIGIR deviated from this principle, along with both of the CHI publications. This is evidence that the CHIIR community appreciates the value of the default search interface design pattern, and understands the benefit of augmenting it rather than replacing it. Those publishing in other venues may be wishing to push the boundaries on what is possible, without considering the impact on the learned human behaviour of searching.

Overall, providing visual enhancements to the search interface was very common within the publications. As not all readers will be familiar with these visualization approaches, they are briefly described in Table 3. Of the 25 publications in this scoping review, only two used non-visual enhancements. Clearly, the research community is embracing the value of leveraging information visualization techniques to enhance search interfaces, even if they do not explicitly comment on this aspect in their publications.

Considering what aspect of the search process was enhanced using visual representations, the SERP was most common followed closely by the workspace. As both of these are fundamentally representing collections of search results, there was overlap in the methods used (highlighting, colour encoding, and node-link representations were most common). Of those that presented visual representations of the query, a variety of approaches were used. Only five search interfaces attempted to use visual representations to enhance the search result items. There remain many opportunities for enhancing further aspects of the search interface with visual representations, especially in the context of individual search results and the query.

Most of the work assessed use well-designed visual approaches (e.g., colour encoding, nesting, highlighting, histograms, timelines, etc.) or textual information that enabled the scrutability of what is being conveyed. However, there were some exceptions where it was difficult to interpret the information provided with respect to the core exploratory search process it was meant to support. Two of these were published in CHIIR, two in CHI, and one each in UIST. Overall, those proposing interfaces to support exploratory search are ensuring that what is provided can indeed be understood in the context of the primary task.

Not surprisingly, many of the proposed approaches are highly interactive; much of the interaction is via simple selection, although a number of methods are using more advanced interaction techniques such as drag-and-drop or using a lasso to draw a closed region to encapsulate items of interest. Of those approaches that are not explicitly interactive, they do update the information provided based on the query and corresponding search results.

Relatively few of the works supported some type of persistence of the interactive efforts the searchers have taken during their exploratory search. Those that provide a results level of persistence directly benefit searchers who delve deeply into the search results for a given query. Those that provide a session level of persistence directly benefit complex search tasks that include the searcher issuing multiple queries. Those that provide a global level of persistence directly benefit cross-session searching. The level of persistence was influenced by the what aspect of the search interface was enhanced; those that made additions to the query, SERP, or result item had a greater opportunity for results or session persistence, whereas those that made additions to the workspace had a greater opportunity for global persistence. Designers of exploratory search interfaces should carefully consider how long the interactivity should persist, and the impacts of this for supporting the searcher.

The publications assessed in this scoping review spanned a wide variety of search contexts, showing the broad reach and scope for designing search interfaces to support exploratory search.

There remain many opportunities for enhancing all aspects of the search interface, using carefully designed visual representations that are scrutable, especially in the context of helping searchers to make sense of their queries and individual search result items. Significant opportunities exist for making the interactivity of such interfaces persist beyond their immediate use, enabling the decision making that went into using the interactive feature to benefit the full search session or even globally across search sessions.

## 6 Discussion & Conclusion

As much of the research on exploratory search interface designs has been guided by intuition about what features might benefit searchers, this has resulted in wide variety of approaches being developed in an ad hoc way. The design principles for exploratory search interfaces presented in this paper provide guidance for future design work, a mechanism for considering the similarities and differences among approaches, and highlighting opportunities for further research directions.

Preserving the default search interface design pattern and including lightweight additions can present a challenge in terms of interface layout limitations, especially in a mobile search context. Further studies on the limits of compactness of the “query box and ten blue links” design pattern (both in desktop and mobile settings) could inform future search interface designs on how much space remains for the lightweight additions and visual representations.

Whether the query, SERP, result item, or workspace serves as the basis for the innovation in the search interface design depends on the motivations and goals of the individual researchers. Developing workspaces features has been popular, recognizing that an important part of the exploratory search processes is saving and analyzing the resources that have been discovered. Extending these

approaches to support global persistence of the interactivity of analyzing the saved resources provides opportunities for enabling and studying cross-session searching.

Many of the approaches used for enhancing the SERP can be applied to workspaces, as both contain collections of search results. An open question is whether the interactivity with a set of search results to enable the discovery of relevant resources has the same value when applied to a workspace and the goal of post-hoc analysis of saved resources. A similar line of thought can be applied to whether such interactivity with the SERP enhancements should persist at the results, session, or global levels.

Notably less work has been done to enhance the query or result items within exploratory search interfaces. There are opportunities for further research on interactive query refinement that leverages visual approaches, and using a compact version of the visual language of the query to explain the relevance of individual search result items. A particular challenge here is scrutability - any visualization used for multiple purposes and presented at multiple levels of granularity would need to be designed carefully to ensure that it remains understandable for all intended purposes.

These design principles have been presented as general principles for exploratory search interface design that are agnostic with respect to the search context. There is space for creating context-specific extensions of these design principles that reflect the unique nature of searching within specific collections, such as academic digital libraries or medical repositories. Similarly, interface-specific extensions may be created to account for the differences between desktop search, mobile search, conversational search, VR/AR search, and other emerging interface technologies.

This work was conducted with a focus on academic studies of exploratory search interfaces. Such studies seldom include input from search interface designers working in industry. Further, those working in industry seldom present the results of their search interface design efforts within academic venues. This disconnect between academia and industry highlights opportunities for refining these design principles based on input from search interface designers working in industry.

With increasing acceptance and adoption of generative AI across a broad range of information-centred settings, a question being asked both in academia and industry is whether generative AI will replace the need to use a search interface. One such example is retrieval-augmented generation, where generative AI is used to provide an answer to a searcher’s query without their need to evaluate or click-through to the search results that provided the source information for the answer [13, 69]. While such approaches appear to be quite effective for look-up search tasks, complex search scenarios require a deeper engagement with the search process than simply viewing a single generated answer. Recent approaches outside of the parameters of the scoping review have proposed the use of generative AI to enhance exploratory search interfaces, include using it to summarize saved resources in a workspace [46] or providing a ChatAI interface alongside a more typical search interface [72]. Within the context of the design principles presented in this paper, these approaches are classified as lightweight additions that are interactive and persistent. There are many opportunities for further work on the use of generative AI to dynamically create visualizations to further enhance the exploratory search process.

The design principles presented here provide a framework for a systematic approach to the design and study of exploratory search interfaces. That is, existing approaches within the specific search context can be evaluated using these design principles and the associated heuristic evaluation framework, enabling the comparison of approaches and identification of opportunities for further research. Moving away from ad hoc designs and toward principle-driven design will enable the search interface design community working within the exploratory search space to highlight where their new contributions are being made, moving the field forward in a systematic way.

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